

S6 — Immunity, Infection and Forensics

6.1 Pathogens and Disease

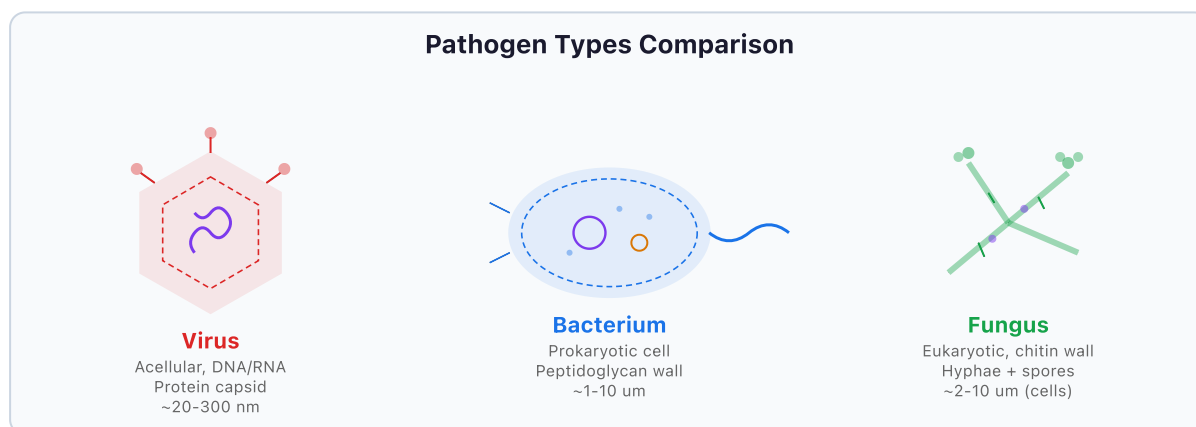
Types of Pathogen

Pathogen: A microorganism that causes disease. Pathogens include bacteria, viruses, fungi, and protoctists (protists). They cause disease by damaging cells directly, producing toxins, or triggering excessive immune responses.

Pathogen Type	Characteristics	Examples
Bacteria	Prokaryotic; cell wall with peptidoglycan; reproduce by binary fission; some produce toxins (exotoxins or endotoxins)	<i>Mycobacterium tuberculosis</i> (TB), <i>Vibrio cholerae</i> (cholera)
Viruses	Acellular; contain DNA or RNA within a protein coat (capsid); obligate intracellular parasites; hijack host cell machinery to replicate	HIV, influenza, SARS-CoV-2
Fungi	Eukaryotic; cell walls of chitin; may be unicellular (yeasts) or multicellular (moulds); reproduce by spores	<i>Candida</i> (thrush), <i>Trichophyton</i> (athlete's foot)
Protoctists	Eukaryotic; diverse group; often transmitted by vectors	<i>Plasmodium</i> (malaria), <i>Trypanosoma</i> (sleeping sickness)

Comparing the Structure of Bacteria and Viruses

Feature	Bacteria	Viruses
Size	About 0.5–5 μm	About 20–300 nm (much smaller)
Cellular?	Prokaryotic cells	Not cells; only replicate inside host cells
Genetic material	Circular DNA free in the cytoplasm (no nucleus); may also have plasmids	DNA or RNA, single- or double-stranded
Outer layers	Cell surface membrane, peptidoglycan cell wall; some have a capsule	Protein coat (capsid); some have a lipid envelope with attachment proteins
Ribosomes / cytoplasm	70S ribosomes in cytoplasm	None
Other structures	May have flagella and pili	Some carry enzymes (e.g. reverse transcriptase in HIV)



Transmission of Disease

Pathogens can be transmitted through various routes:

- **Direct contact:** Physical touch, sexual contact, or exchange of body fluids (e.g., HIV, sexually transmitted infections).
- **Airborne (droplet):** Coughing, sneezing, or breathing releases pathogen-laden droplets (e.g., TB, influenza).
- **Waterborne:** Contaminated water supplies (e.g., cholera, typhoid).
- **Foodborne:** Contaminated food (e.g., *Salmonella*, *E. coli*).

- **Vector-borne:** An intermediate organism transmits the pathogen (e.g., *Anopheles* mosquito transmits *Plasmodium*).
- **Fomites:** Contaminated surfaces or objects.

How TB and HIV Infect Human Cells

Tuberculosis (*Mycobacterium tuberculosis*): bacteria in inhaled droplets reach the lungs and are engulfed by macrophages, but survive inside them (their thick, waxy cell wall resists digestion and they stop lysosomes fusing with the phagosome). In **primary infection** an inflammatory response walls the infected macrophages off in **tubercles**, where bacteria can stay dormant for years. If the immune system is weakened, active (post-primary) TB develops: the bacteria multiply and destroy lung tissue. Symptoms include a persistent cough (sometimes with blood), fever, night sweats, loss of appetite and weight loss. Without treatment the loss of lung function and spread to other organs can cause death.

HIV (Human Immunodeficiency Virus): a retrovirus spread in body fluids. Its glycoprotein (gp120) binds to CD4 receptors on **T helper cells** (and macrophages); the envelope fuses with the membrane and the RNA and enzymes enter. **Reverse transcriptase** makes DNA from the viral RNA, and **integrase** inserts it into the host genome, where it may stay latent. When active, new viruses are made and bud off, destroying T helper cells. Stages: an acute flu-like phase, a long chronic phase with few symptoms while T helper numbers fall, then **AIDS**, when T helper numbers are so low that opportunistic infections (e.g. TB, pneumonia) and some cancers develop, weight loss and dementia may occur, and these secondary infections cause death.

6.2 The Immune System

Non-Specific (Innate) Immunity

Non-specific defences provide immediate, general protection against all pathogens. They include:

- **Physical barriers:** Skin (keratinised epithelium), mucous membranes, cilia in airways, stomach acid (pH 1-2 denatures proteins).
- **Chemical barriers:** Lysozyme in tears and saliva (breaks down bacterial cell walls), sebum on skin (lowers pH), antimicrobial peptides.
- **Skin and gut flora:** Harmless commensal bacteria on the skin and in the gut compete with pathogens for nutrients and space, and some secrete substances (e.g. lactic acid) that inhibit

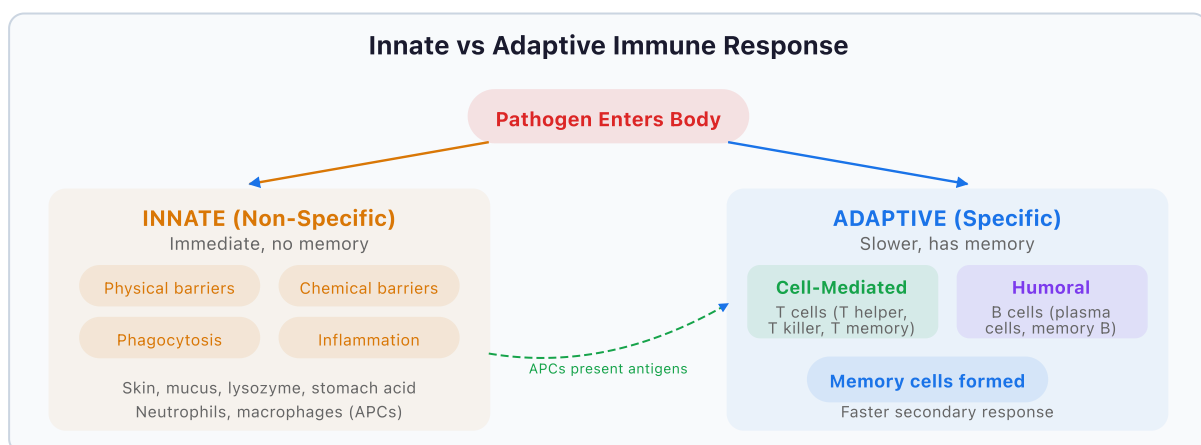
pathogens. Broad-spectrum antibiotics can kill this flora and allow infections such as *C. difficile* to take hold.

- **Phagocytosis:** Neutrophils and macrophages recognise, engulf, and destroy pathogens in phagosomes. Lysosomes fuse with the phagosome and release hydrolytic enzymes that digest the pathogen. Macrophages also act as antigen-presenting cells.
- **Inflammation:** Damaged cells release histamine, causing vasodilation and increased capillary permeability. This brings more white blood cells and plasma (containing antibodies and complement) to the infected area.
- **Interferon:** Cells infected by viruses release interferons, which bind to nearby cells and inhibit viral replication in them (e.g. by stopping viral protein synthesis), slowing the spread of the virus. Interferons also activate other immune cells.

Specific (Adaptive) Immunity

Antigen: A molecule (usually a protein or glycoprotein) on the surface of a cell that can be recognised by the immune system. Foreign antigens on pathogens trigger an immune response. Self-antigens on the body's own cells are normally tolerated.

The specific immune response involves lymphocytes and is characterised by specificity (each lymphocyte responds to one specific antigen) and immunological memory.



Cell-Mediated Immunity (T Cells)

T lymphocytes mature in the thymus and respond to antigens presented on the surface of infected cells by antigen-presenting cells (APCs). Types of T cell include:

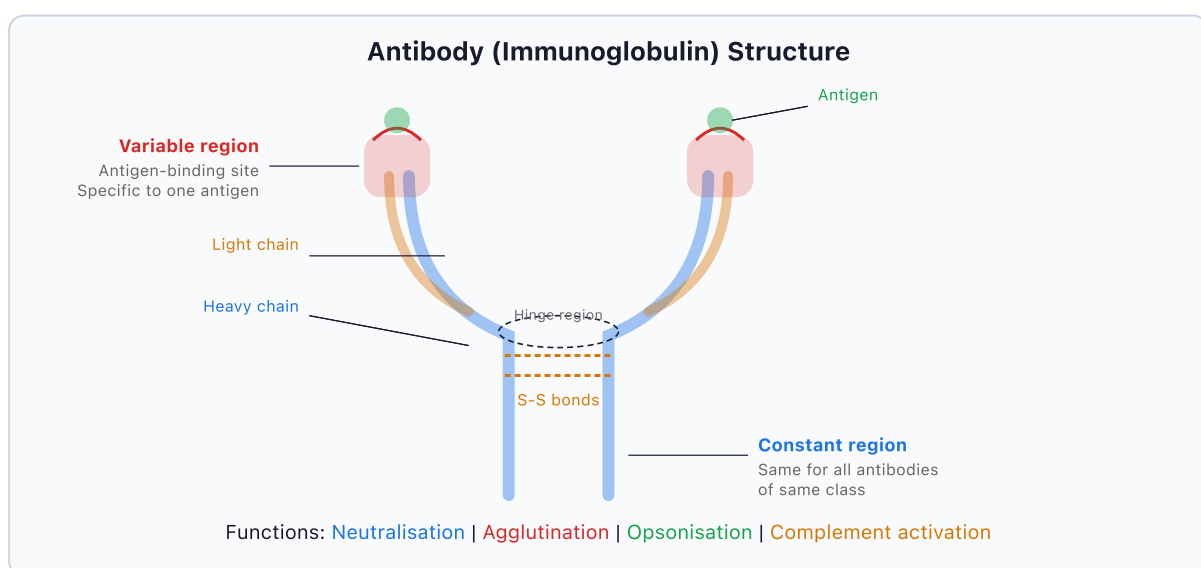
- **T helper cells (T_H):** Recognise antigens on APCs and release cytokines that activate B cells, cytotoxic T cells, and macrophages. Essential for coordinating the immune response.
- **Cytotoxic T cells (T_C / killer T cells):** Directly kill infected body cells by releasing perforin (creates pores in cell membrane) and granzymes (trigger apoptosis).
- **Regulatory T cells (T_{reg}):** Suppress the immune response after infection is cleared, preventing autoimmune reactions.
- **Memory T cells:** Remain in the body long-term and enable a faster, stronger secondary response upon re-infection.

Humoral Immunity (B Cells)

B lymphocytes mature in the bone marrow and produce antibodies. When a B cell encounters its specific antigen and is activated by T helper cells, it undergoes clonal expansion (rapid mitotic division) and differentiates into:

- **B effector cells → plasma cells:** B effector cells differentiate into plasma cells, short-lived cells that secrete large quantities of antibodies (immunoglobulins) into the blood and tissue fluid. Each plasma cell produces one specific antibody.
- **Memory B cells:** Long-lived cells that persist after infection and enable a rapid secondary immune response (more antibodies produced more quickly upon re-exposure).

Antibody Structure and Function



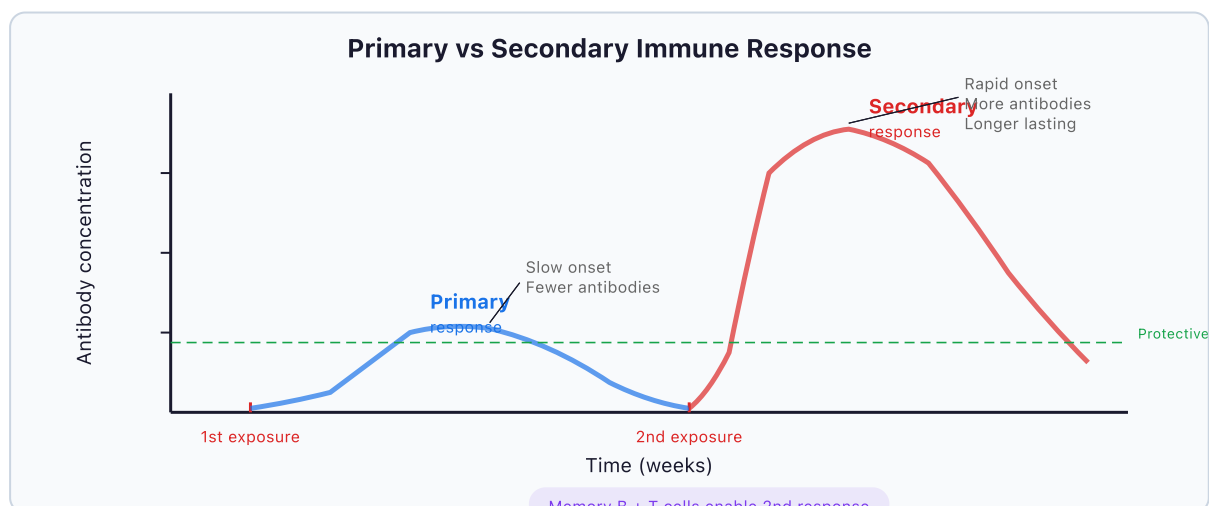
Antibodies are Y-shaped glycoproteins with two heavy chains and two light chains held together by disulfide bonds. The variable regions at the tips of the Y are complementary in shape to a specific antigen (the antigen-binding site). The constant region determines the class of antibody and how it functions. Antibodies neutralise pathogens by:

- **Neutralisation:** Binding to toxins or viral attachment proteins, preventing them from interacting with host cells.
- **Agglutination:** Cross-linking pathogens together, making them easier for phagocytes to engulf.
- **Opsonisation:** Coating pathogens with antibodies, tagging them for phagocytosis.
- **Complement activation:** Triggering the complement system, which forms membrane attack complexes (pores) that lyse pathogen cells.

One Gene, More Than One Protein

One gene can give rise to more than one protein through **post-transcriptional changes to mRNA**. The primary transcript (pre-mRNA) contains exons and introns. Introns are removed by splicing, and the exons can be joined in different combinations (**alternative splicing**), sometimes with some exons also removed. Each different mature mRNA is translated into a different polypeptide, so one gene can code for several proteins (e.g. the membrane-bound and secreted forms of the same antibody).

6.3 Vaccination and Antibiotics



Vaccination

Vaccination: The administration of a weakened, killed, or subunit form of a pathogen (or its toxin) to stimulate an immune response and the production of memory cells, without causing the disease. This provides long-term protection against future infection.

Types of vaccine include:

Type	How It Works	Example	Advantages / Disadvantages
Live attenuated	Weakened pathogen that replicates but does not cause disease	MMR, oral polio, BCG	Strong, long-lasting immunity; may revert to virulent form (rare); unsuitable for immunocompromised
Inactivated	Killed pathogen; cannot replicate	Inactivated polio, rabies	Safer; weaker immune response; may need boosters
Subunit / recombinant	Purified antigen (e.g., surface protein) only	Hepatitis B, HPV	Very safe (no pathogen at all); may need adjuvants to enhance response
Toxoid	Inactivated toxin	Tetanus, diphtheria	Targets the toxin rather than the pathogen itself
mRNA	mRNA instructs host cells to produce a pathogen protein, triggering immunity	COVID-19 (Pfizer, Moderna)	Rapid development; no live pathogen; requires cold storage

Herd immunity occurs when a sufficient proportion of a population is immune (through vaccination or prior infection) that transmission is greatly reduced, protecting unvaccinated individuals (e.g., newborns, immunocompromised) indirectly. The threshold for herd immunity depends on the pathogen's basic reproduction number (R_0) — for measles ($R_0 \sim 12-18$), approximately 95% vaccination coverage is needed, while for polio ($R_0 \sim 5-7$), about 80-85% suffices.

Antigenic variation: Some pathogens (e.g., influenza, HIV, trypanosomes) can change their surface antigens through mutation or recombination. This means that memory cells from a previous infection or vaccination may not recognise the new variant, so the individual is susceptible again. This is why new influenza vaccines are developed annually and why HIV is so difficult to vaccinate against.

The evolutionary race: hosts evolve better defences, and pathogens with mutations that let them evade these defences survive and multiply, so each side acts as a selection pressure on the other. Evasion mechanisms support this theory. **HIV** has a high mutation rate in its antigen genes (causing antigenic variation), destroys the T helper cells that coordinate the immune response, can remain latent inside host cells, and interferes with antigen presentation by infected cells. **M. tuberculosis** survives inside macrophages, stops lysosomes fusing with them, suppresses antigen presentation, and stays dormant in tubercles.

Exam Tip: Distinguish between active immunity (body produces its own antibodies and memory cells, either naturally through infection or artificially through vaccination — long-lasting) and passive immunity (antibodies received from an external source, either naturally via placenta/breast milk or artificially via injection — short-lived because no memory cells are formed).

Antibiotics

Antibiotics are substances produced naturally by microorganisms (or synthetic derivatives) that kill (**bactericidal**, e.g., penicillin) or inhibit the growth of (**bacteriostatic**, e.g., tetracycline) bacteria. They work by targeting bacterial structures or processes that differ from human (eukaryotic) cells:

Target	Antibiotic Examples	Mechanism
Cell wall synthesis	Penicillin, amoxicillin, vancomycin	Inhibit cross-linking of peptidoglycan; weakened wall leads to osmotic lysis
Protein synthesis (70S ribosomes)	Tetracycline, erythromycin, chloramphenicol	Block translation at bacterial 70S ribosomes (different from eukaryotic 80S)
DNA replication / transcription	Fluoroquinolones (ciprofloxacin), rifampicin	Inhibit bacterial DNA gyrase or RNA polymerase
Metabolic pathways	Sulfonamides, trimethoprim	Block folic acid synthesis (bacteria must synthesise it; humans obtain it from diet)

Antibiotics are ineffective against viruses because viruses lack cell walls, ribosomes, and their own metabolic pathways — they replicate using host cell machinery, so any drug targeting the virus would

also damage host cells. Antiviral drugs exist but work by different mechanisms (e.g., blocking viral attachment, inhibiting viral enzymes like reverse transcriptase or protease).

Bacteriostatic vs bactericidal: bacteriostatic antibiotics inhibit bacterial growth and division (e.g. by blocking protein synthesis), and the body's immune system then destroys the bacteria; bactericidal antibiotics kill the bacteria directly (e.g. by weakening the cell wall so the cells burst). Some antibiotics are bacteriostatic at low concentration and bactericidal at high concentration.

Core Practical 15: Investigate the effect of different antibiotics on bacteria. Using aseptic technique (disinfect the bench, work near a Bunsen flame, flame the loop and bottle necks, lift the Petri dish lid as little as possible), spread a bacterial culture evenly over an agar plate. Place paper discs soaked in different antibiotics (and a control disc without antibiotic) on the plate, tape the lid (do not seal it fully) and incubate at 25°C. Measure the diameter of each clear zone of inhibition and calculate its area (πr^2); a larger zone shows greater effectiveness.

Antibiotic Resistance

Antibiotic resistance arises through natural selection. Random mutations in bacterial DNA may confer resistance (e.g., producing beta-lactamase enzymes that break down penicillin). In the presence of antibiotics, resistant bacteria survive and reproduce while susceptible bacteria are killed, increasing the frequency of resistance alleles. Resistance genes can spread between bacteria via horizontal gene transfer (conjugation, transformation, transduction). MRSA (methicillin-resistant *Staphylococcus aureus*) is a well-known example of a resistant pathogen.

Measures to reduce antibiotic resistance include: completing prescribed courses, not prescribing antibiotics for viral infections, using narrow-spectrum antibiotics where possible, infection control in hospitals, restricting antibiotic use in agriculture, and research into new antibiotics and alternative treatments (e.g., bacteriophage therapy).

Hospital Acquired Infections

Hospital acquired infections (HAIs), such as MRSA and *C. difficile*, are infections caught while in hospital. Contributory causes include: many ill or immunocompromised patients close together; invasive procedures, wounds and catheters; high antibiotic use selecting for resistant strains; and spread by staff hands, equipment and surfaces. Understanding these causes has led to **codes of practice**: on **antibiotic prescription** (prescribe only when needed, use narrow-spectrum antibiotics after identifying the bacterium, avoid overuse of broad-spectrum antibiotics, ensure the course is completed) and on

hospital practice for infection prevention and control (hand washing or alcohol gel between patients, bare below the elbows, screening and isolating infected patients, thorough cleaning and disinfection of wards and equipment, sterilising instruments).

6.4 Forensic Biology

Time of Death

Forensic scientists estimate time of death using multiple methods:

- **Body temperature:** After death, body temperature decreases toward ambient temperature (algor mortis). The rate depends on body mass, clothing, air temperature, and environmental conditions.
- **Rigor mortis:** Stiffening of muscles beginning 2-4 hours after death (actin-myosin cross-bridges cannot be broken without ATP) and fading after 36-48 hours as muscle proteins denature.
- **Livor mortis:** Pooling of blood in lowest parts of the body due to gravity, causing purple/red discolouration. Fixed after ~12 hours.
- **Decomposition stages:** Fresh, bloat, active decay, advanced decay, dry remains. Influenced by temperature, moisture, and oxygen availability.
- **Forensic entomology:** The succession of insect species colonising a body follows a predictable pattern. Blowflies are typically first to arrive. The life cycle stage of insects (egg, larva, pupa, adult) can indicate time since death.
- **Stage of succession:** A body is colonised by a predictable succession of organisms (bacteria, then blowflies, then flesh flies, beetles and later mites), each group changing the body so that it suits the next. The species present indicate the time since death.

DNA Profiling (Genetic Fingerprinting)

DNA profiling: A technique that produces a pattern of DNA fragments unique to an individual (except identical twins), based on variations in short tandem repeats (STRs) — repetitive sequences of DNA that vary in number between individuals at specific loci.

The process of DNA profiling:

1. **DNA extraction:** DNA is isolated from a biological sample (blood, saliva, hair, semen, skin cells).

2. **PCR amplification:** Specific STR loci are amplified using the polymerase chain reaction to produce millions of copies from a tiny sample.
3. **Electrophoresis:** DNA fragments are separated by size using gel electrophoresis (smaller fragments travel further). Fragments may be visualised using fluorescent probes.
4. **Analysis:** The resulting banding pattern is compared between samples. Multiple loci (typically 10-13) are analysed to give extremely high statistical confidence.

The Polymerase Chain Reaction (PCR)

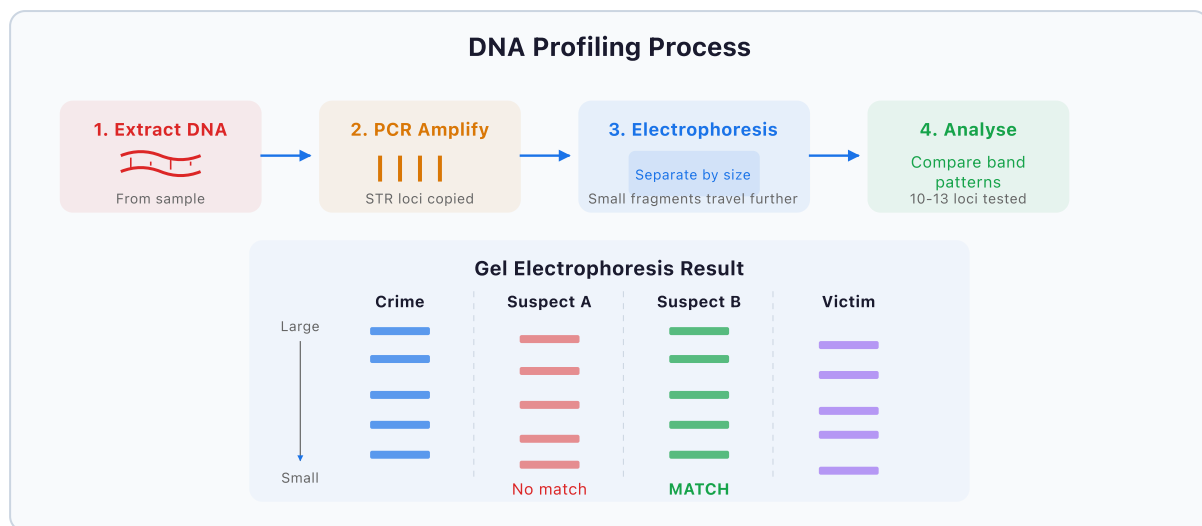
PCR amplifies DNA from a tiny sample. The reaction mixture contains the DNA sample, **primers** (short single-stranded DNA complementary to the start of the region to be copied), free DNA nucleotides and a heat-stable **DNA polymerase** (Taq polymerase), in a buffer. Each cycle has three steps:

1. **Denaturation (about 95°C):** hydrogen bonds between the strands break, separating the DNA into single strands.
2. **Annealing (about 55–60°C):** primers bind by complementary base pairing to the ends of the target sequence.
3. **Extension (about 72°C):** DNA polymerase adds nucleotides to the primers, making complementary strands.

The amount of target DNA doubles each cycle, so after n cycles there are 2^n copies of each starting molecule (e.g. 30 cycles give about 10^9 copies).

Core Practical 14: Use gel electrophoresis to separate DNA fragments of different length.

DNA samples (with a loading dye) are placed in wells at one end of an agarose gel covered in buffer, and a voltage is applied. DNA is negatively charged (phosphate groups), so it moves towards the positive electrode (anode); shorter fragments move faster through the gel and travel further. A DNA ladder of fragments of known length is run alongside for comparison, and the DNA is then made visible with a stain (e.g. methylene blue) or a fluorescent dye viewed under UV light.



Applications of DNA profiling include criminal investigations (matching suspects to crime scenes), paternity testing, identifying victims of disasters, immigration cases, and evolutionary/conservation studies. DNA profiles are also used to determine **genetic relationships between organisms**, both animals and plants (e.g. confirming parentage in breeding programmes and identifying plant varieties): the more bands two profiles share, the more closely related the individuals are. Ethical considerations include privacy, the national DNA database, potential for misuse, and the risk of contamination leading to false matches.

Exam Tip: In questions about DNA profiling, explain each step clearly and emphasise why STRs (not coding regions) are used — they are highly variable between individuals. Remember that DNA profiling shows probability of a match, not certainty. Contamination and degradation of samples are key limitations.

6.5 Decomposition and Succession

Decomposition

Decomposition is the breakdown of dead organic matter by microorganisms (bacteria and fungi) and detritivores (e.g., earthworms, woodlice). Decomposers secrete extracellular enzymes onto organic matter, digesting it externally (saprotrophic nutrition), then absorbing the soluble products. Decomposition recycles nutrients (carbon, nitrogen, phosphorus) back to the soil and atmosphere, making them available for new plant growth.

Micro-organisms and the recycling of carbon: as bacteria and fungi decompose dead organic matter they respire the products of digestion, releasing carbon dioxide (and methane in anaerobic conditions) into the atmosphere. This CO_2 is then taken up by plants in photosynthesis, so decomposers return carbon locked in dead organisms to the carbon cycle.

Factors affecting the rate of decomposition include temperature (affects enzyme activity), oxygen availability (aerobic decomposition is faster), moisture (required for enzyme activity and microbial growth), pH, and the nature of the material (soft tissue decomposes faster than bone or lignin).

Ecological Succession

Succession: The gradual and directional change in the species composition of a community over time, eventually leading to a stable climax community. Primary succession begins on bare, previously uncolonised substrate; secondary succession occurs after disturbance of an existing community.

Pioneer species (e.g., lichens, mosses) are the first to colonise bare rock. They are adapted to harsh conditions and help create soil by weathering rock and adding organic matter when they die. As soil develops, more complex plant communities can establish, each seral stage modifying the environment to make it more suitable for the next stage. Eventually, a climax community is reached (determined by climate and soil type), which is relatively stable and self-sustaining.